

CYBER SECURITY INCIDENT RESPONSE & SOC ANALYSIS

Incident Response Laboratory Report

Prepared By: Lakshmi Subhaja Kari

Degree: B.Tech – Cybersecurity

Role Simulated: SOC Analyst

Project Year: 2026

Classification: Controlled Educational Laboratory Exercise

1. Executive Summary

This project simulates the role of a Security Operations Center (SOC) Analyst investigating a suspected authentication-related security incident on a Windows endpoint.

The investigation focused on repeated failed authentication attempts recorded in the Windows Security Event Log as **Event ID 4625**.

Three relevant failed-logon events were identified on **07 October 2026**. The events were investigated to determine the affected account, logon type, failure reason, status codes, workstation and source address.

The investigation identified the affected account as **soc-test** on the Windows endpoint **DESKTOP-IC8QE5D**. The source network address was **::1**, which represents the local Windows system.

The incident response lifecycle was completed through:

Detection → Investigation → Classification → Containment → Recovery → Lessons Learned

During containment, the **soc-test** account was temporarily disabled. After investigation, the account was re-enabled to complete the recovery process.

This was a **controlled cybersecurity laboratory simulation**. The evidence collected during the exercise does not establish a real external attacker, confirmed brute-force attack, malware infection or data exfiltration.

2. Project Objectives

The main objectives of this project were:

- Detect suspicious authentication activity.
- Analyze Windows Security Event Logs.
- Investigate Windows Event ID 4625.
- Identify the affected account.
- Identify the authentication failure reason.
- Analyze the source of the authentication request.
- Classify the incident.
- Perform incident containment.
- Perform recovery.
- Create an incident timeline.
- Identify the root cause.
- Document lessons learned.
- Recommend preventive security controls.

3. Lab Environment

Component	Details
Operating System	Windows 11
Hostname	DESKTOP-IC8QE5D
Test Account	soc-test
Log Source	Windows Security Event Log
Primary Event	Event ID 4625
Logon Type	2 – Interactive
Source Address	:::1 – localhost
Environment	Controlled Cybersecurity Laboratory
Role Simulated	SOC Analyst

4. Incident Scenario

A simulated security incident was created by generating multiple failed authentication attempts against a dedicated test account named **soc-test**.

The SOC Analyst was required to investigate the authentication activity and determine:

1. What happened?
2. Which account was affected?
3. Why did the authentication fail?
4. Where did the authentication request originate?
5. Was repeated activity present?
6. What containment action should be taken?
7. How should the system be recovered?
8. What security controls could prevent similar incidents?

The exercise was conducted in a controlled laboratory environment.

5. Incident Detection

During the investigation, Windows Security Event ID **4625** was identified.

Event ID 4625 represents a failed logon attempt.

Three relevant events were identified:

Date	Time	Activity
07-Oct-2026	3:16:55 PM	First failed authentication
07-Oct-2026	3:21:32 PM	Second failed authentication
07-Oct-2026	5:49:12 PM	Third failed authentication

Evidence

****Figure 1: Windows Security Event ID**